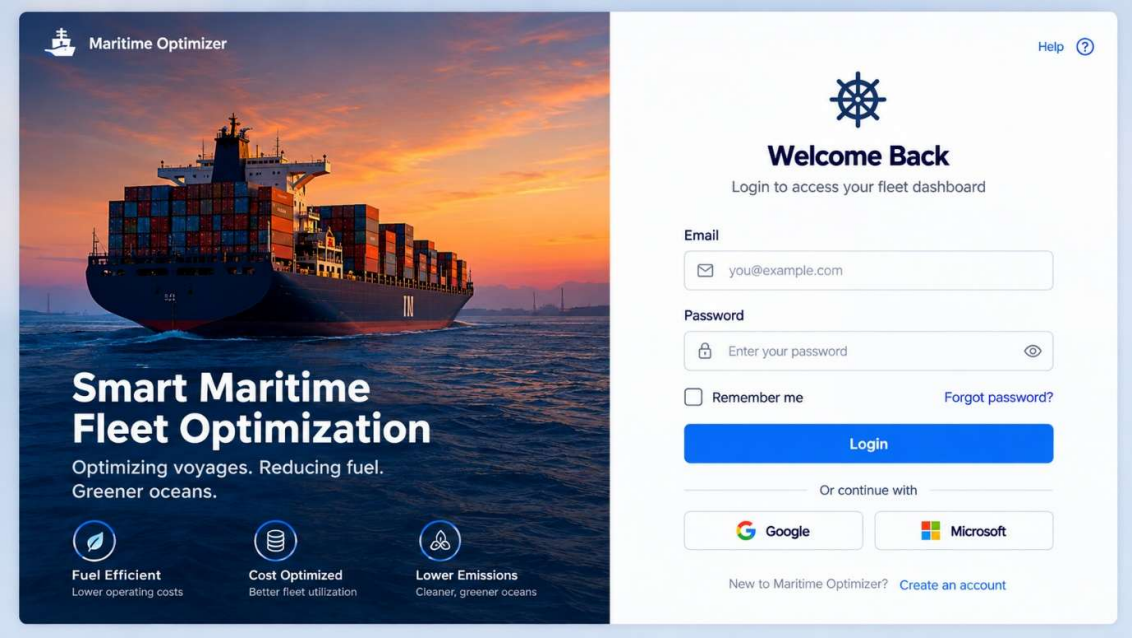
Frame 1 — Login Screen is generated above.

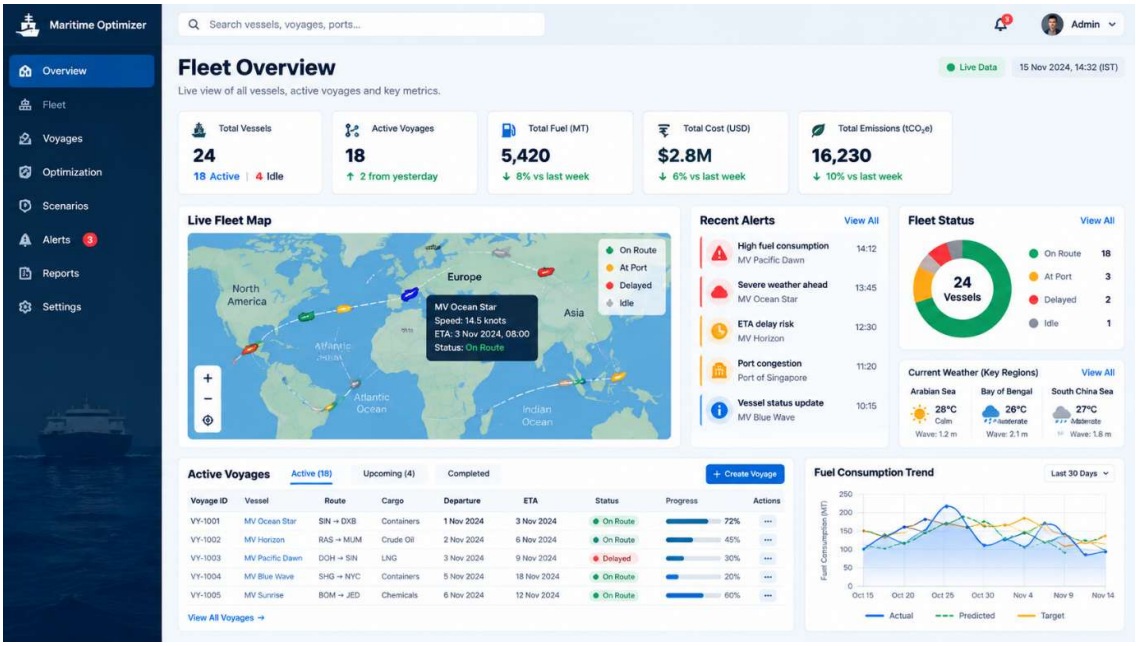


Frame 2 — Fleet Overview Dashboard

This is the **main home screen** after login.

It shows:

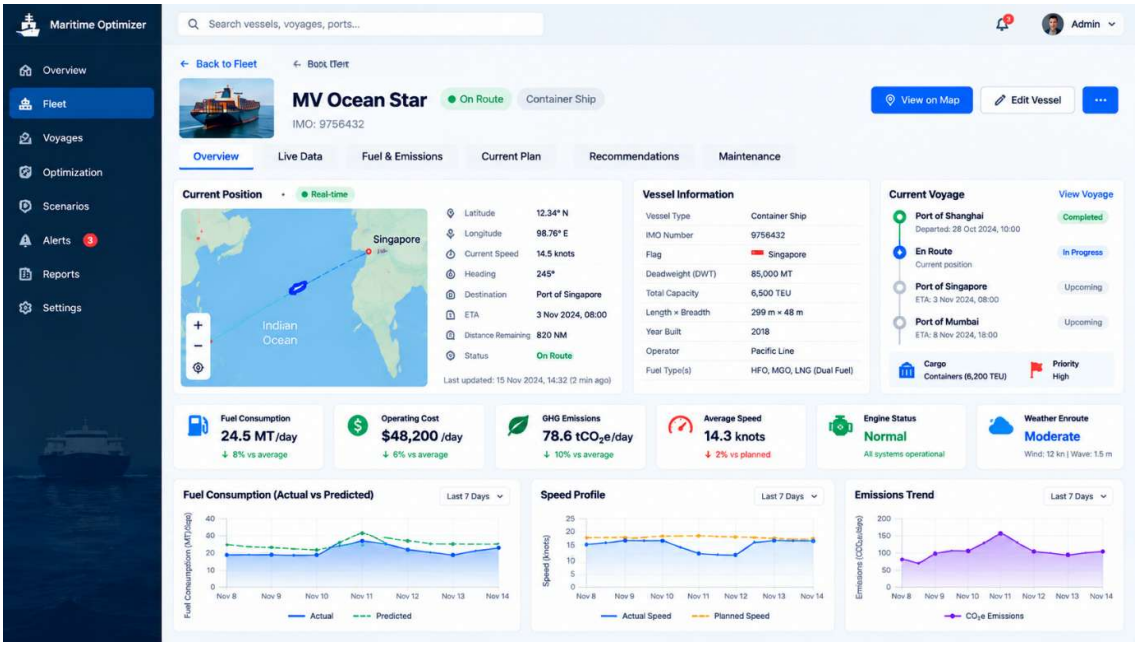
- Live fleet map
- Total vessels & active voyages
- Fuel consumption
- Operating cost
- GHG emissions
- Vessel status
- Current weather
- Alerts
- Active voyages
- Fuel prediction vs actual trend



Frame 3 — Vessel Details

This screen is focused on **one vessel**, showing:

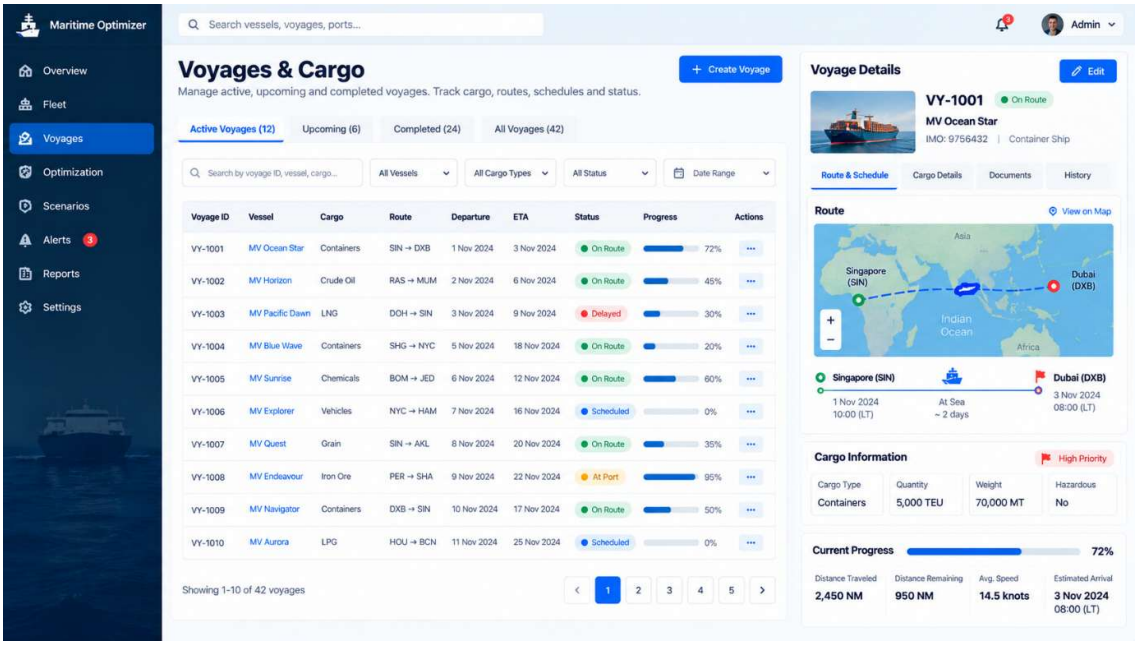
- Real-time position & route
- Vessel specifications
- Current voyage
- Cargo and priority
- Fuel consumption
- Operating cost
- GHG emissions
- Speed vs planned speed
- Actual vs predicted fuel
- Weather conditions
- Engine status
- Current plan & recommendations



Frame 4 — Voyages & Cargo

This screen manages all **voyages and cargo operations**:

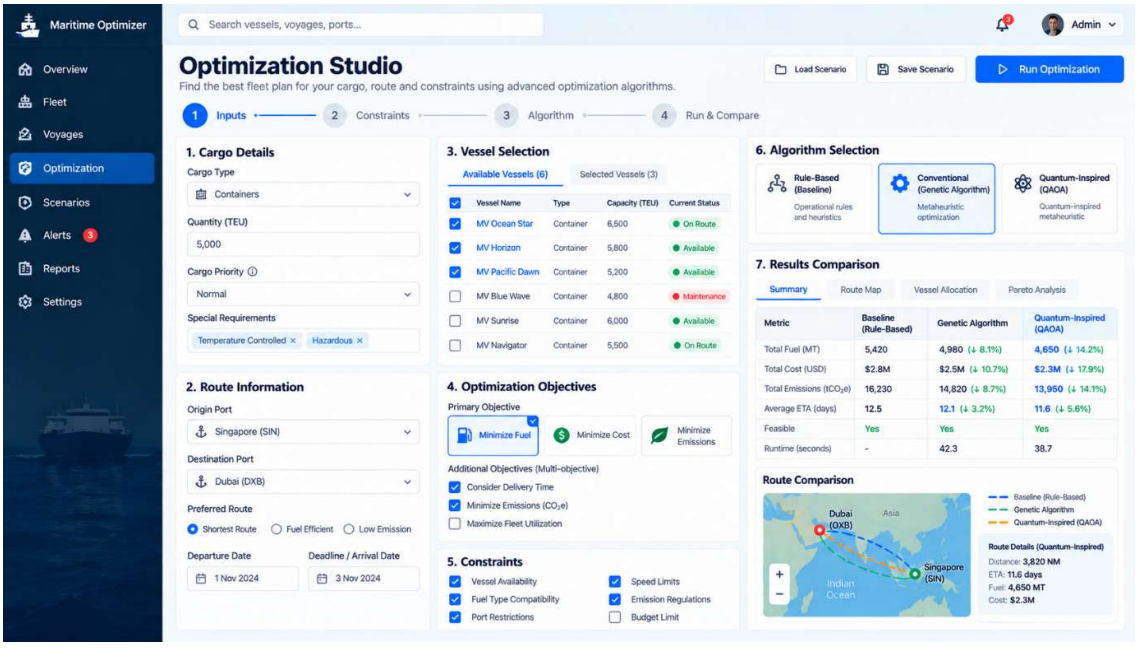
- Active / upcoming / completed voyages
- Vessel and cargo filtering
- Route and ETA
- Voyage status
- Progress tracking
- Route map
- Cargo quantity & weight
- Priority
- Distance travelled/remaining
- Estimated arrival



Frame 5 — Optimization Studio

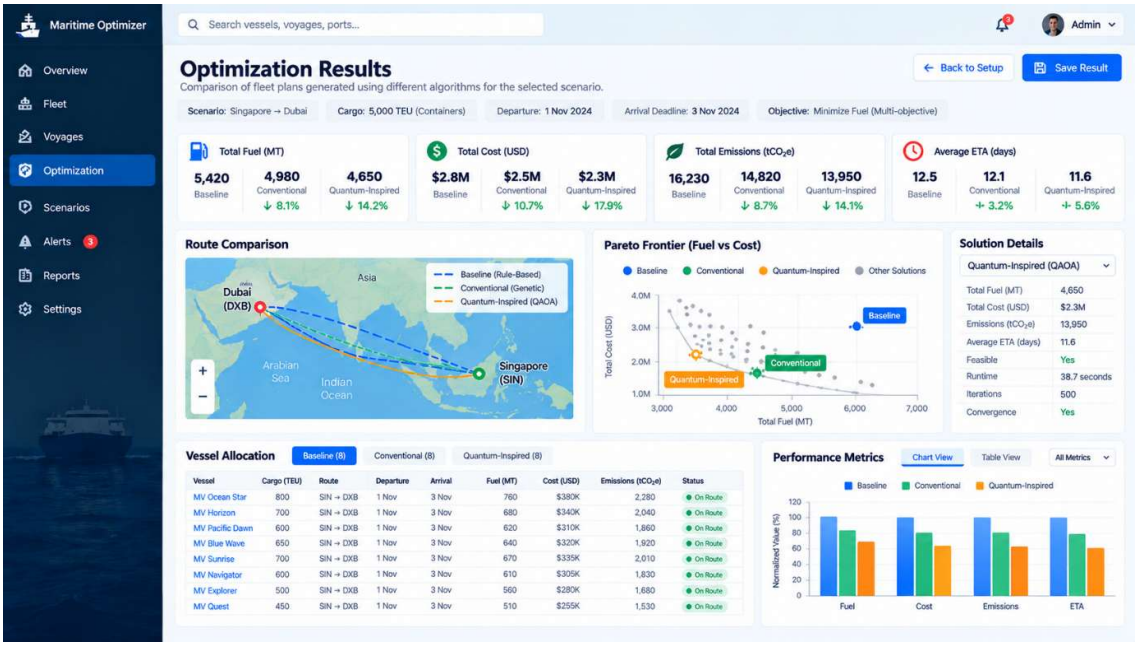
This is the **core screen of the project**.
Here the operator:

- Selects cargo
- Selects vessels
- Defines origin/destination
- Sets deadline
- Chooses optimization objectives
- Sets operational constraints
- Selects:
  - Rule-Based Baseline
  - Conventional Algorithm
  - Quantum-Inspired Algorithm
- Runs optimization
- Compares fuel, cost, GHG, ETA
- Views route differences
- Checks feasibility



Frame 6 — Optimization Results
This is where the system **proves what the optimizer achieved**.
It shows:

- Baseline vs Conventional vs Quantum-Inspired
- Fuel comparison
- Cost comparison
- GHG comparison
- ETA comparison
- Route comparison
- **Pareto Frontier**
- Selected solution details
- Vessel allocation
- Runtime & iterations
- Performance charts



Frame 7 — Scenario Simulator

This is the **"What if?" screen**.

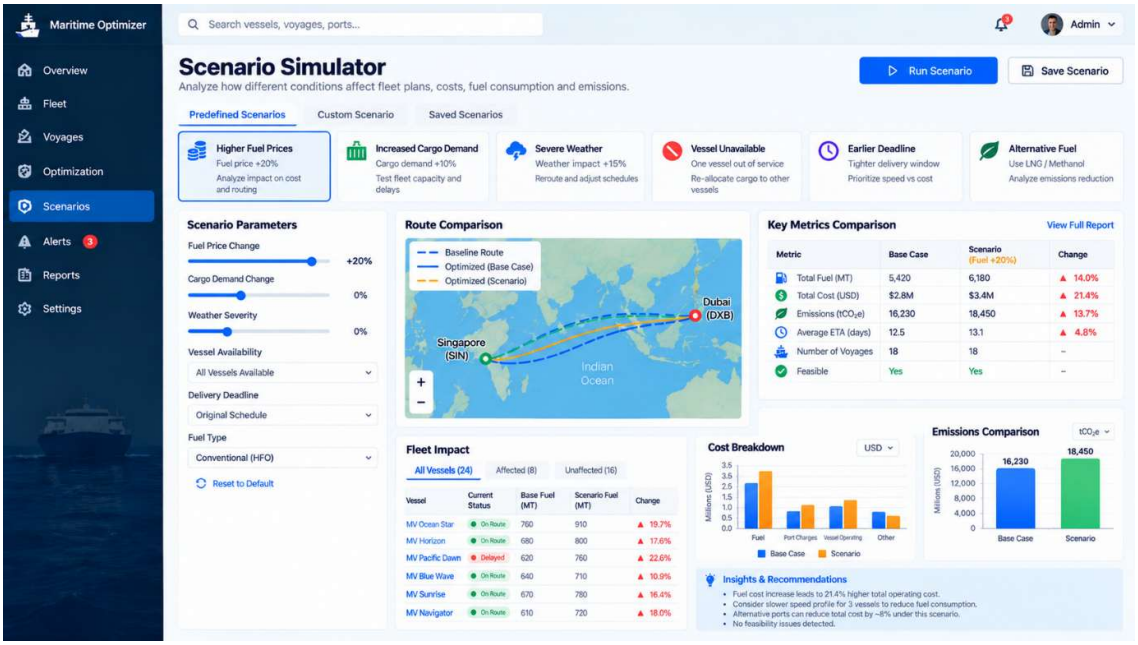
You can change:

- Fuel price
- Cargo demand
- Weather severity
- Vessel availability
- Delivery deadline
- Fuel type

Then the system recalculates:

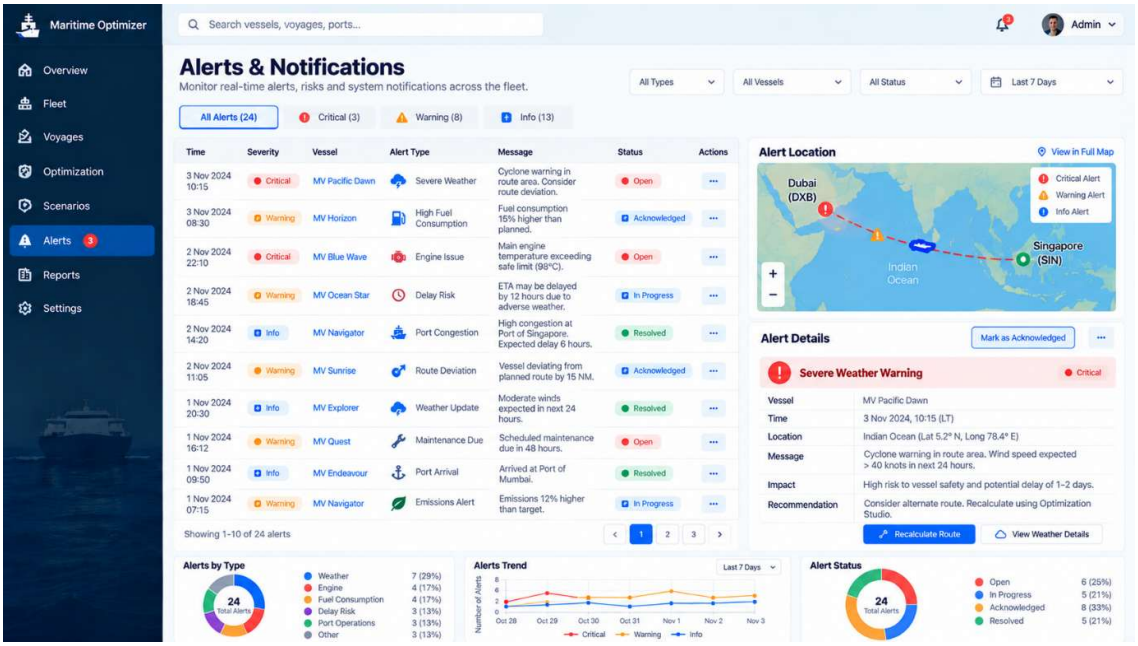
Route → Fuel → Cost → GHG → ETA → Fleet allocation

It also explains the impact and gives recommendations



Frame 8 — Real-Time Alerts & Notifications

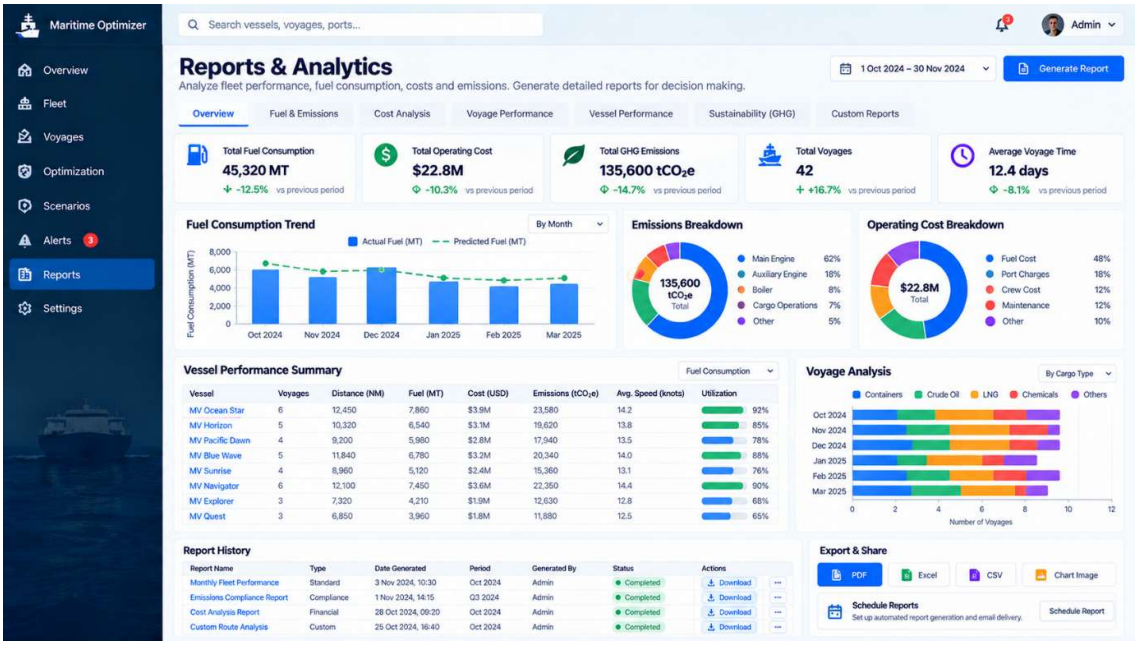
This screen monitors **live operational risks** across the fleet, with critical/warning/info alerts, vessel and time filters, alert locations on the map, detailed impact/recommendation panels, and actions such as acknowledging alerts or recalculating a route.



Frame 9 — Reports & Analytics

This is the **management and performance screen**, showing:

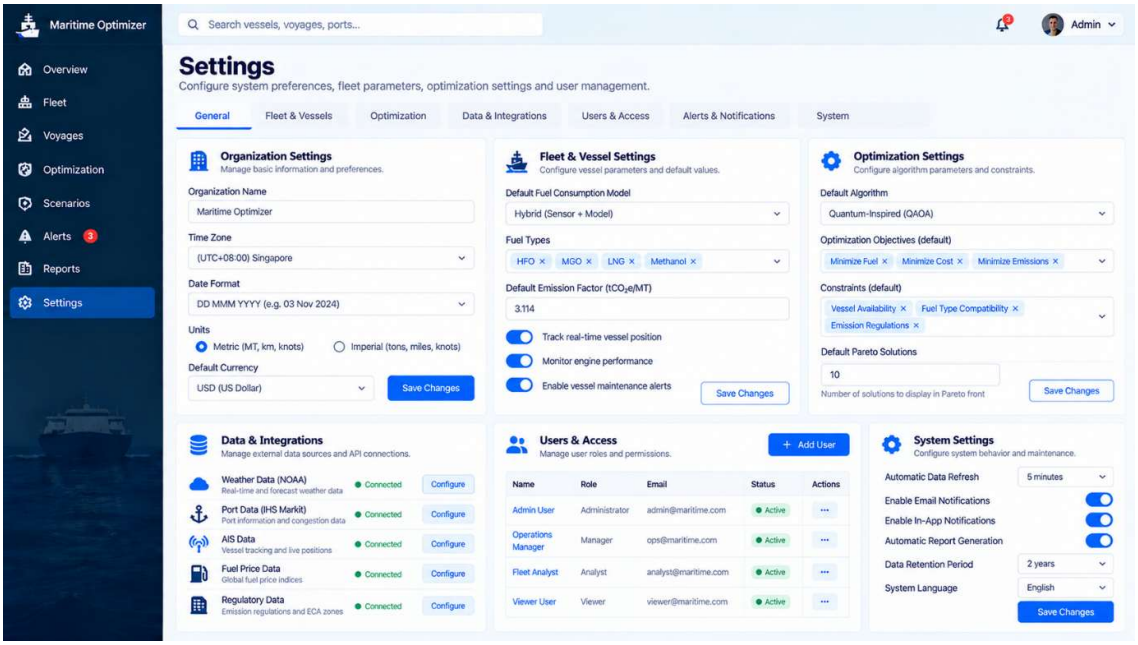
- Fuel consumption trends
- Operating costs
- GHG emissions
- Voyage performance
- Vessel-wise performance
- Fleet utilization
- Cost/emission breakdowns
- Historical reports
- PDF / Excel / CSV export
- Scheduled reports



Frame 10 — Settings & System Configuration

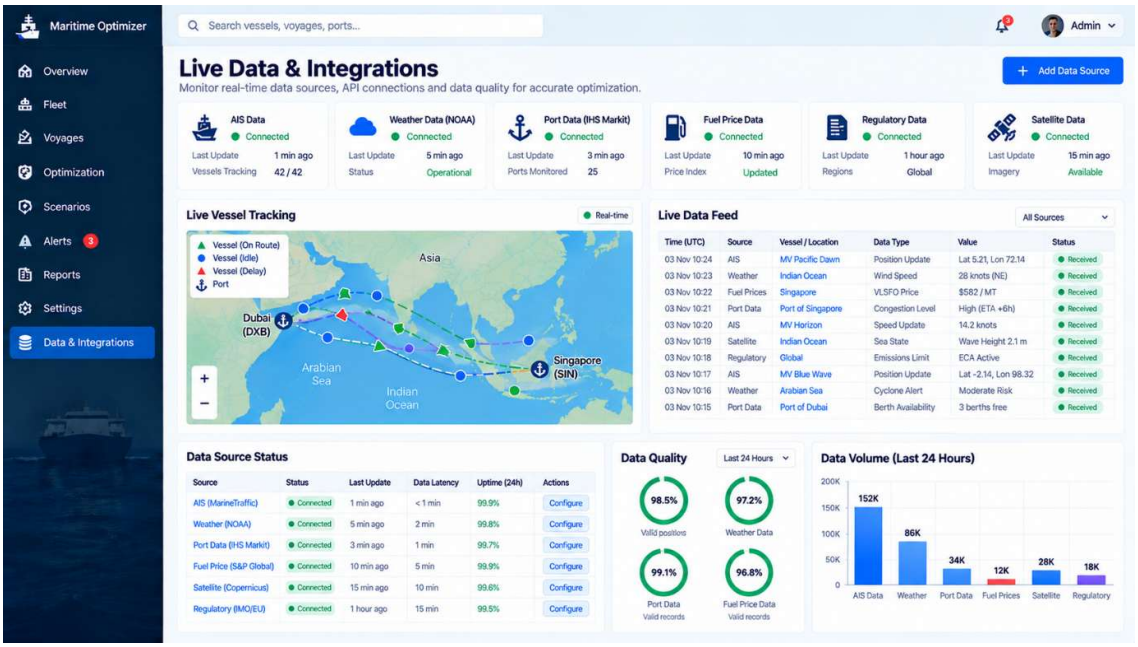
This screen controls the platform itself:

- Fleet & vessel defaults
- Optimization parameters
- Fuel types & emission factors
- Data/API integrations
- User roles & permissions
- Alert settings
- System refresh & notifications
- Model/algorithm defaults



Frame 11 — Live Data & Integrations
This is the **data-control center** of the platform.
It shows:

- AIS data
- Weather data
- Port data
- Fuel prices
- Regulatory data
- Satellite data
- Live vessel tracking
- Incoming data feed
- Source health & latency
- Data quality
- Data volume

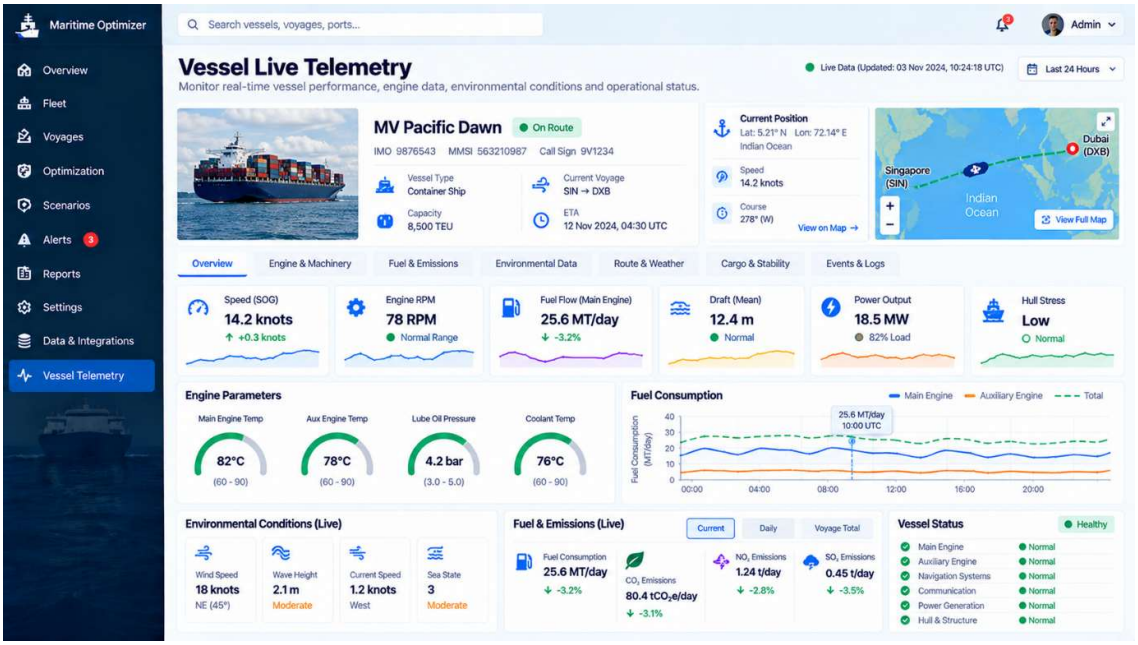


Frame 12 — Vessel Live Telemetry

This screen connects the **real-time data layer directly to the vessel**:

- Live position, speed & course
- Engine RPM and temperatures
- Fuel flow
- Draft & power output
- Hull stress
- Environmental conditions
- Fuel consumption
- CO_2 / NO_2 / SO_2 emissions
- Engine health
- Real-time vessel status

This is the screen that makes the **"live data → prediction → optimization"** architecture visually clear.

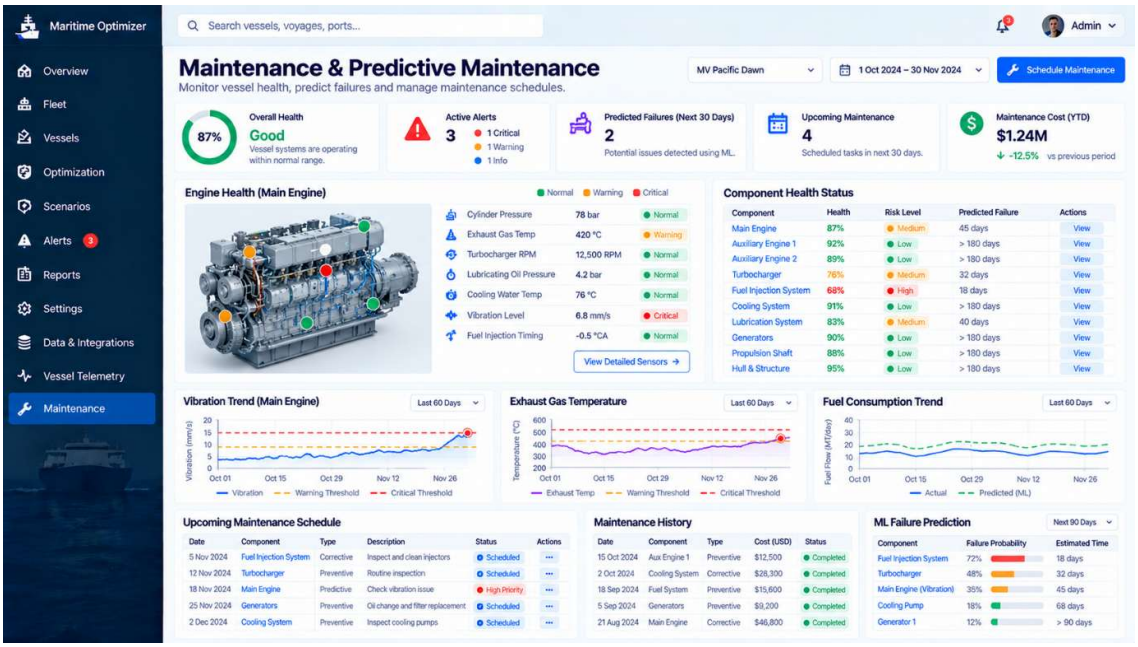


Frame 13 — Maintenance & Predictive Maintenance

This screen monitors **vessel health and maintenance risk**:

- Overall vessel health
- Active maintenance alerts
- ML-predicted failures
- Engine/component health
- Sensor readings
- Vibration & temperature trends
- Fuel-consumption trends
- Upcoming maintenance
- Maintenance history
- Failure probability and estimated failure time

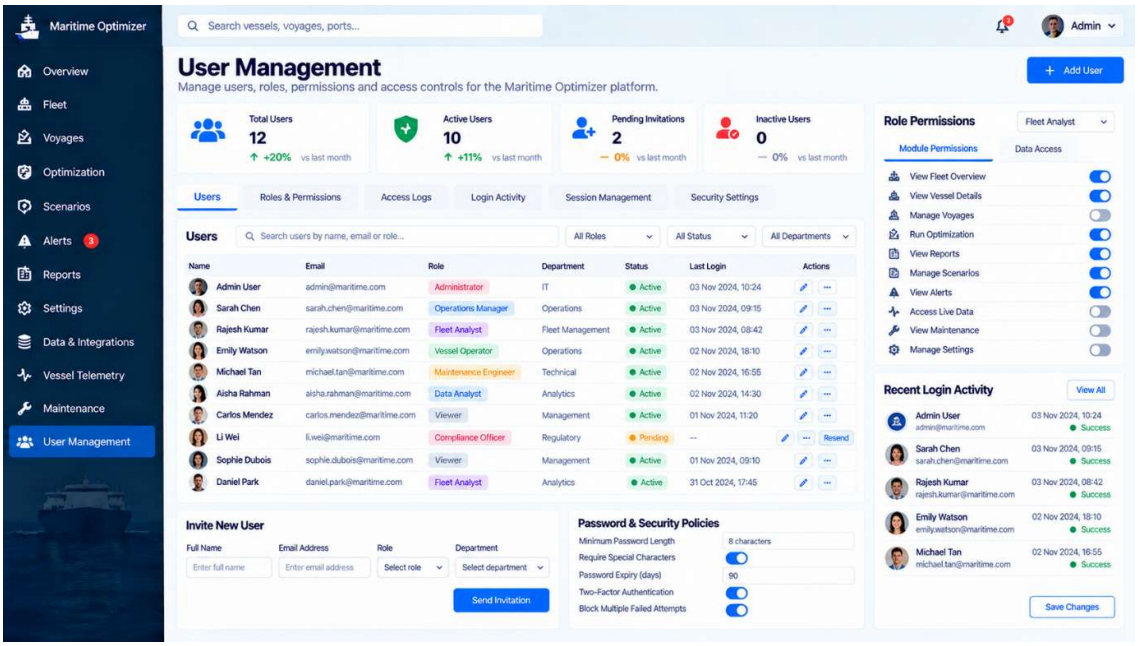
This can later connect directly to the **real-time telemetry + ML pipeline**.



Frame 14 — User Management

This screen handles **access control and security**:

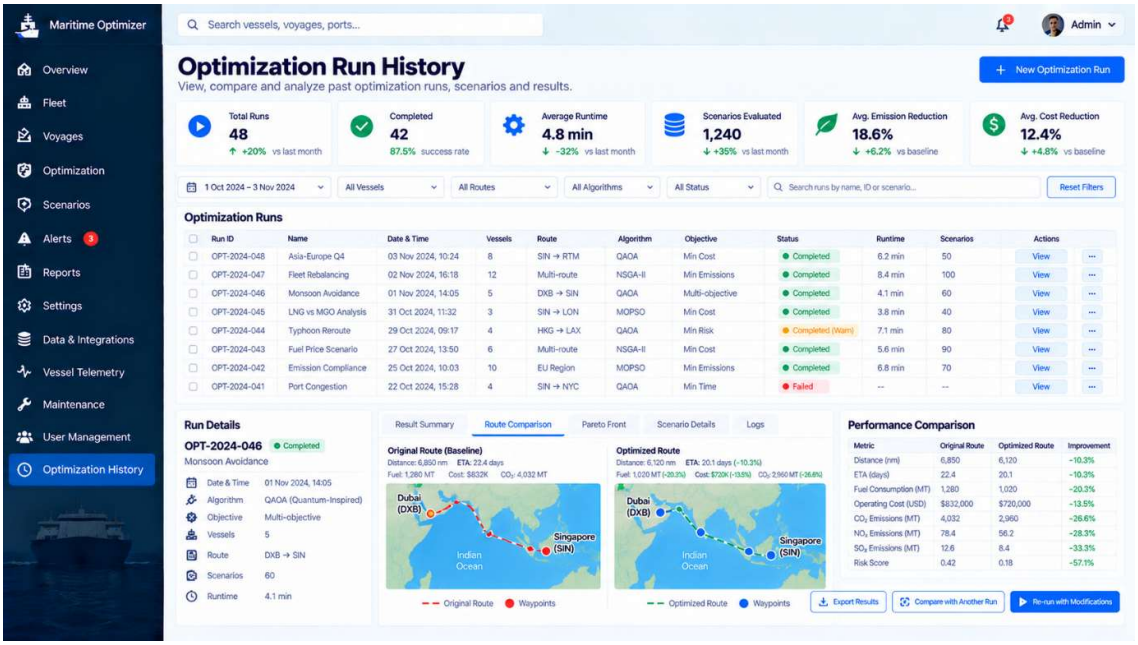
- User directory
- Roles & permissions
- Module-level access
- Data access
- Login activity
- Session management
- User invitations
- Password/security policies



Frame 15 — Optimization Run History

This screen keeps a **complete history of every optimization run**:

- Run ID & scenario
- Algorithm used
- Vessels & routes
- Runtime
- Success/failure status
- Fuel/cost/emission improvements
- Original vs optimized route
- Pareto results
- Detailed run logs
- Compare previous runs
- Export results
- Re-run with modified parameters



Frame 16 — System Health & Monitoring
This is the **technical control room** for your platform:

- Backend/API health
- ML service
- Optimization engine
- PostgreSQL
- Redis
- WebSockets
- Data-source connectivity
- CPU / memory / disk / network
- API latency
- System uptime
- Recent system events
- Infrastructure alert

